cryptic or warning coloration are advantageous or adaptive, and indeed were inclined to dismiss the whole idea of adaptation contemptuously as mere teleological speculation. The Lamarckians and the Vitalists rejected the idea of natural selection as too materialistic and as giving insufficient weight to will and effort and other psychological forces. The early Mendelians, fascinated by the discovery of genetic units (gene-differences) with large effects (such as hornless-ness in cattle, or albinism in many vertebrates), wanted to make mutation responsible for positive evolutionary change, and assigned to natural selection only the essentially negative function of getting rid of harmful variants. Many of them rejected the idea of slow biological improvement and gradual evolution in favour of stepwise mutational changes of large extent. The position was finally complicated by the fact that the biometricians, preoccupied with the gradual variation so often found in nature, wanted to deny the importance of Mendelism, and confused the heritable variations of character produced by changes in the genes with the nonheritable ones produced by changes in the environment.

Eventually, however, these various contradictions were reconciled. Lamarckian and vitalistic explanations were ruled out when it was shown that acquired characters, whether impressed by the environment or resulting from use or individual effort, were never inherited. The advance of genetics showed that large mutations were rarer and of far less biological importance than those of small extent, and that, apparently, continuous evolutionary change could be, and often was, brought about by the accumulation of numerous small discontinuous mutations under the guidance of natural selection. And finally R. A. Fisher in 1930 made it clear that the fact of heredity being particulate—dependent on distinct self-reproducing units or genes, each of which could mutate into new self-reproducing forms—and the further fact that most mutants are recessive, at once got rid of the major difficulties that beset Darwin, who accepted the current view of blending heredity—the view that characters and the entities that determined them were commingled into a single blend when crossed. This would imply that any new character would be progressively diluted by crossing in each generation, and would make its establishment in the stock difficult. But a particulate genetic mechanism in which most mutants are recessive makes it possible for new mutants to be stored indefinitely in the constitution, and for new combinations of new and old genes to be formed, ready to be utilized by selection when conditions are favourable. Natural selection was seen, not as involving the sharp alternatives of life or death, but as the result of the differential survival of variants; and it was established that even slight advantages, of one-half of one percent or less, could have important evolutionary effects.